

Full length article

Phytogetic diet enhances the growth and disease resistance of the black-spotted frog, *Pelophylax nigromaculatus*, by stabilizing intestinal homeostasis and boosting innate immunity

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ABSTRACT

Amphibian aquaculture has rapidly expanded due to the increasing global demand for animal proteins. Among cultured species, black-spotted frog (*Pelophylax nigromaculatus*) has gained attention because of its high economic and market value. However, frequent disease outbreaks under intensive farming conditions considerably hinder its sustainable growth. Recently, plant-derived functional additives have emerged as promising alternatives to enhance disease resistance in aquaculture due to their immunostimulatory, antioxidative, and gut microbiota-modulating activities. This study evaluated the effects of different concentrations of a herbal supplement (HS) containing 0.32 % ginsenosides and 20 % ginseng polysaccharides on *P. nigromaculatus* growth performance and disease resistance over a 60-day feeding trial. Compared to the controls, frogs fed HS-supplemented diets showed significantly higher specific growth and weight gain rates ($P < 0.05$). Serum antioxidant enzyme activity was significantly elevated in the HS groups ($P < 0.05$). Intestinal barrier-related gene (zona occludens 1 and 2 and claudin 3 and 5) levels were upregulated, and antimicrobial peptide gene *OL-CATH1* levels were significantly increased, whereas pro-inflammatory cytokine gene (interleukin-10 and -12) levels were markedly down-regulated ($P < 0.05$), suggesting enhanced innate immunity without systemic inflammation. Moreover, 16S rRNA sequencing revealed a notable increase in beneficial gut bacteria abundance and reduction in pathogenic species abundance in HS-treated frogs, indicating a favourable shift in intestinal microbiota composition. In the pathogen challenge test with *Aeromonas hydrophila*, HS-fed frogs exhibited significantly higher survival rates than the controls ($P < 0.05$). Validation trial revealed that *Lactococcus lactis* dietary supplementation improved the frog growth performance and gut microbial stability. Overall, HS enhanced the growth performance and disease resistance by modulating the gut microbiota, boosting the antioxidant capacity, and promoting the innate immune responses of *P. nigromaculatus*. Our findings suggest 0.04 % HS as a safe, eco-friendly, and sustainable functional additive for amphibian aquaculture.

1. Introduction

Recently, increasing global demand for animal proteins has accelerated aquaculture development. However, modern farming techniques, such as intensive farming, heterotrophic mixed systems, and biofloc technology, have increased the disease prevalence in fish. Although traditional antibiotic treatments temporarily suppress the spread of

diseases, they present significant challenges, including drug residues, emergence of antibiotic-resistant pathogens, and environmental pollution. Therefore, identification of effective antibiotic alternatives is critical for the sustainable development of aquaculture industry [1]. Nutritional status is a key factor determining the disease resistance of aquatic animals. Adjustment of nutrient intake for enhanced immunity is a healthy, eco-friendly, and non-toxic approach to effectively promote sustainable aquaculture development [2]. Ongoing research on the

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Glossary	
ALT	alanine aminotransferase
AST	aspartate aminotransferase
CAT	catalase
CLD	claudin
HS	herbal supplement
IL	interleukin
SGR	specific growth rate
SOD	superoxide dismutase
SR	survival rate
WGR	weight gain rate
ZO-1	zona occludens-1

relationship between the nutrition and immunity of aquatic animals is mainly focused on essential nutrients, green functional feed additives, and fermented feed. Among these, herbal and plant bioactive compounds are promising feed additives enhancing the immune responses and infectious disease resistance of fish [3]. Various medicinal herbs and dietary feed supplements have been studied for their potential to counteract various pathogens and activate immunopoietic cells [4]. For example, mulberry leaf extract enhances the growth, gastrointestinal health, and liver function of giant salamanders [5]. Similarly, rosmarinic acid effectively mitigates the adverse effects of high soybean meal diets by attenuating intestinal inflammation and oxidative stress in bullfrogs (*Lithobates catesbeianus*) [6]. *Spirulina* protects against cadmium-induced toxicity in African clawed frogs (*Xenopus laevis*) [7]. Use of dill (*Anethum graveolens*) powder as a dietary supplement enhances the lysozyme activity and antioxidant enzyme expression in the liver of African catfish (*Clarias gariepinus*), thereby reducing their susceptibility to *Edwardsiella* infections [8]. Bioactive compounds commonly found in plant extracts, such as polyphenols and flavonoids, modulate the immune-related cytokines, including interleukin (IL)-1 β , tumour necrosis factor- α , and IL-10, and antioxidant defence systems across various fish species [9]. Plant pigments, such as lycopene and β -carotene, also exhibit significant antimicrobial activity against *Pseudomonas aeruginosa*, primarily by

modulating the Toll-like receptor pathways and suppressing pro-inflammatory cytokine expression, thereby enhancing the innate immune responses in *Channa punctata* (spotted snakehead) [10]. Additionally, plant-derived compounds promote the growth and reproductive performance of fish by regulating the growth-related factors, such as growth hormone and insulin-like growth factor-1, and reproductive hormones, such as gonadotropin-releasing hormone and follicle-stimulating hormone- β [11]. Collectively, these reports suggest that medicinal plants and their derivatives holistically improve the growth and disease resistance of aquatic animals via multiple mechanisms, including modulation of immune- and antioxidant-related gene expression, activation of enzymatic antioxidant defence, improvement of gut health, and inhibition of pathogenic colonisation.

Ginseng (*Panax ginseng*), an *Araliaceae* family member, contains various bioactive compounds, such as ginsenosides, amino acids, alkaloids, phenolic compounds, polysaccharides, peptides, and vitamins [12]. It is widely used in traditional Chinese medicine to treat human fatigue. Animal and in vitro studies have revealed that ginseng enhances immunoregulation by promoting cytokine synthesis and improving the B-cell- and T-cell-mediated immune responses [13]. Ginseng extract, derived from the perennial herb *Panax ginseng*, contains various active compounds, including ginsenosides, polysaccharides, volatile oils, amino acids, and vitamins. Ginseng extract is frequently used as a natural plant-based additive in aquaculture feed because of its ability to enhance growth performance, disease resistance, and metabolism [14].

Current research on the application of ginseng extract to aquatic species encompasses a dozen species, including *Litopenaeus vannamei* [15], *Penaeus indicus* [16], *Penaeus monodon* [17], crucian carp (*Carrasius auratus*) [18], large yellow croaker (*Larimichthys crocea*) [19], Nile tilapia (*Oreochromis niloticus*) [20], African catfish (*Clarias gariepinus*) [21], and olive flounder (*Paralichthys olivaceus*) [22]. Ginseng extract significantly enhances the growth performance of experimental fish, improving their resistance to bacterial infections and boosting their antioxidant capacity, thereby showing substantial application potential. However, research on amphibians remains limited, with studies primarily focusing on American bullfrogs (*Rana catesbeiana*) and Siberian frogs (*Rana amurensis*) [23].

Pelophylax nigromaculatus, one of the most widely distributed amphibians in China, exhibits a delicate taste and high nutritional and market value, attracting significant attention from both researchers and farmers [24]. Although stocking offers substantial economic benefits to farmers, inappropriate stocking density often leads to disease outbreaks. Li [25] conducted a comprehensive virome analysis of 100 *P. nigromaculatus* individuals and identified 28 high-quality viral sequences from at least 11 viral families. These reports underscore the complexity of viral infections caused by this species, which, if left unchecked, lead to severe morbidity and mortality in frog populations. Bacterial infections pose daunting challenges. For instance, *Elizabethkingia miricola*, a gram-negative bacterium, is associated with devastating conditions, such as meningitis and “twisted head syndrome” in frogs [26]. This pathogen is also associated with urinary tract and congenital joint infections in humans, posing zoonotic risks [13].

Current disease prevention measures for *P. nigromaculatus* focus primarily on controlling individual pathogens, with limited success in addressing the broader issue of disease susceptibility. Existing interventions often lack a systematic approach to enhance the frog innate immunity, which is crucial for general defence against various pathogens. The lack of such comprehensive strategies exacerbates the vulnerability of *P. nigromaculatus*, hindering its sustainable farming [13]. Therefore, holistic approaches balancing both growth and health are essential to ensure the long-term viability of *P. nigromaculatus* farming. Enhancing the innate immunity of these amphibians is not only a theoretical goal but also a practical necessity for reducing disease prevalence, minimising the use of chemical treatments, and mitigating the risks associated with zoonotic pathogens. Identification of green and safe methods to boost immune resilience will improve the productivity

Table 1
Compositions and nutrient contents of the experimental diets (% dry matter).

Ingredients	Group			
	C	T1	T2	T3
Fish meal	4	4	4	4
Chicken meal	25	25	25	25
Soybean meal	30	30	30	30
Canola meal	15	15	15	15
Flour	19.95	19.95	19.95	19.95
Soybean oil	3	3	3	3
Premix	1	1	1	1
Monocalcium phosphate	1.5	1.5	1.5	1.5
Choline chloride	0.5	0.5	0.5	0.5
Antioxidant	0.02	0.02	0.02	0.02
Mold inhibitor	0.03	0.03	0.03	0.03
Herbal supplement	0	0.02	0.04	0.08
Total	100	100.02	100.04	100.08
Nutritional composition				
Crude protein	40.23	40.06	40.47	40.17
Crude fat	7	7.09	7.28	7.11
Crude ash	12.5	12.4	12.58	12.11

Each kg of premix contains the following: copper, vitamin A = 15000 IU, vitamin D3 = 3000 IU, vitamin E = 200 mg, vitamin K3 = 5 mg, vitamin B1 = 10 mg, vitamin B2 = 20 mg, vitamin B6 = 15 mg, vitamin B12 = 0.04 mg, folic acid = 5 mg, niacin = 50 mg, biotin = 0.2 mg, pantothenic acid = 30 mg, copper sulphate = 10 mg, zinc sulphate = 80 mg, manganese sulphate = 60 mg, potassium iodide = 1.5 mg, sodium selenite = 0.2 mg, lysine = 2 g, and methionine = 1 g.

and promote the environmentally sustainable and ethically responsible farming of frogs [27].

In this study, we investigated the effects of different levels of a herbal supplement (HS; mixture of ginsenosides and ginseng polysaccharides) on the growth performance, gut microbiota, and disease resistance of *P. nigromaculatus*. Our findings offer practical guidance for HS application as a functional feed additive to enhance the productivity and promote the sustainable farming of *P. nigromaculatus*.

2. Materials and methods

2.1. Experimental diets

All ingredients used in this study were provided by Hunan Micro-organism & Herb Biological Technology Co., Ltd. (Xiangtan, China). HS contained 0.32 % ginsenosides (Rg1:Re:Rb1 = 1:1:1) and 20 % ginseng polysaccharides. Ginsenosides were extracted from ginseng stems and leaves, as described by Zhou [28], whereas ginseng polysaccharides were obtained from the by-products of ginseng extract using the hot water extraction method described by Qi [29]. Subsequently, HS was incorporated into the basal diet at 0, 0.02, 0.04, and 0.08 % (per kg of feed) [8]. The other ingredients were finely ground, thoroughly mixed, and processed using a pelletising machine into four types of iso-nitrogenous and isoenergetic extruded pellets with a uniform diameter of 1.00 mm. The pellets were air-dried under natural conditions, vacuum-sealed, and stored at -20°C . All feed ingredients and formulations are listed in Table 1.

2.2. Frogs and experimental procedure

In total, 2000 juvenile frogs (initial weight: 12 ± 0.5 g; healthy and active) were used in this study. Following two-week acclimation in outdoor experimental ponds, the frogs were randomly assigned to four treatment groups, with three replicates per group and 150 individuals per replicate. Experimental ponds were divided into 12 sections (dimensions: $4 \times 1.5 \times 0.6$ m each) using mesh dividers. During the 60-day trial, the frogs were cultured in microflow water systems. Experimental diets were manually administered twice daily (07:00 and 17:00) at a feeding rate of 3 % of body weight [30]. Biweekly sampling and weighing were performed to adjust the feeding rate as necessary.

2.3. Growth parameters

At the beginning and end of the experimental period, 30 frogs were randomly selected from each group and fasted for 24 h, and their body weights were measured. Mortality in each group was continuously recorded throughout the experiment. Specific growth rate (SGR), weight gain rate (WGR), and survival rate (SR) were calculated using the following formulae [31]:

$$\text{SR, \%} = N_t/N_0 \times 100$$

$$\text{WGR, \%} = 100 \times (W_t - W_0)/W_0;$$

$$\text{SGR, \%}/\text{d} = (\ln W_t - \ln W_0) \times 100/t$$

where, W_0 and W_t are the initial and final body weights (g), respectively, N_t is the final number of fish, and N_0 is the initial number of fish.

2.4. Determination of serum biochemical indices

According to the guidelines of the Biomedical Research Ethics Committee of Hunan Agricultural University (Changsha, China), all samples from each population were anaesthetised in the laboratory before the experiment via immersion in MS-222 (Coolaber, Beijing, China) at 200 mg/L for 5 min. Thirty frogs were randomly selected from each group, and their blood samples were collected from the popliteal

Table 2

Primer sequences used for real-time fluorescence quantification polymerase chain reaction (PCR).

Gene	Sequences (5'–3')	Efficiency (%)
ZO-1	F: TTGAAGAGCCGCTCTGTGCTA R: GCCCTTGTTCGATTAAATCCCTG	94.5
ZO-2	F: CTCCATTGTAATTCAGATGTGCTT R: TGCTTGGAACAACCATTTGGACCC	96.8
CLD3	F: ATGTCATGAGGACTTGAGATATTAG R: AAAGGCAGTCACCCCTCCACATA	92.7
CLD5	F: AACATCGTGGTGGCTCAGGTCATCT R: GGTTCTTCACCTTGCTCCCTG	95.2
OL-CATH1	F: GGGGCTCTGCACTCCTTGAGTGGC R: GAGGCATTCTGTCTCTTTATT	98.0
IL10	F: TGACAGTGTTCCTTTGTTATTTT R: CTTTGACACACTGCTCTTACT	92.0
IL12	F: CCCTACAGCAACAGGATTCGAGGG R: TGTAATGGTGACGATGGAGCCC	99.0
NF- κ B	F: GAATGGAGACACGCCCTCCACCTT R: TCAAGGATAGTAGGGTCTGCTC	100.0
β -actin	F: TGGCTTTGGACCTGTGTCTGATG R: GCGATGTGAATCTGTCTCGACTG	93.5

Note: ZO-1, zonula occludens-1 (tight junction protein 1); ZO-2, zonula occludens-2-like isoform X2 (tight junction protein 2); CLD3, claudin-3; CLD5, claudin-5; OL-CATH1, *Odorrana livida* cathelicidin family antimicrobial peptide 1; IL10, interleukin-10; IL12, interleukin-12; NF- κ B, nuclear factor-kappa B.

artery using syringes. Blood samples of every three frogs were pooled into a single centrifuge tube and stored at 4°C overnight. The samples were centrifuged at 3500 rpm for 5 min to separate the serum. Then, alanine aminotransferase (ALT), aspartate aminotransferase (AST), superoxide dismutase (SOD), and catalase (CAT) activities were measured using commercial assay kits (Nanjing Jiancheng Bioengineering Institute, Nanjing, China).

2.5. RNA extraction and quantitative reverse transcription-polymerase chain reaction (RT-qPCR)

Intestinal and liver samples were collected from each group and stored at -80°C for RNA extraction. Total RNA was extracted using the TransZol Up Plus RNA Extraction Kit (Beijing TransGen Biotech Co., Ltd.). RNA concentration and quality were assessed using the K5800 microspectrophotometer (Beijing Kaiuo Technology Development Co., Ltd.). RNA was reverse-transcribed into cDNA using the RevertAid First Stand cDNA Synthesis Kit (Thermo Fisher Scientific). qRT-PCR was performed using the ChamQ Universal SYBR qPCR Master Mix (Vazyme Biotech Co., Ltd.). All target genes and primers used are listed in Table 2. PCR amplification efficiency of each primer pair was calculated using a linear regression model [32]. The amplification efficiency was 92–100 %, indicating that all primers exhibited good amplification performance. Relative mRNA expression levels of genes were calculated using the $2^{-\Delta\Delta\text{CT}}$ method.

2.6. 16S rRNA amplicon sequencing

Intestinal samples were collected from the hindgut of six frogs in each group and sent to Guangzhou Gene Denovo Biotechnology Co., Ltd (Guangzhou, China), for 16S rRNA sequencing. Genomic DNA was extracted from the samples, and conserved regions of rDNA were amplified using specific primers with barcodes. The PCR products were purified via gel extraction and quantified using the QuantiFluor fluorometer. Purified amplification products were mixed in equal proportions, ligated with sequencing adapters, and used to construct the sequencing libraries. Sequencing was performed using the Illumina PE250 platform [33].

After obtaining the raw reads, low-quality reads were filtered, and paired-end reads were assembled into tags. The tags were further filtered to generate clean tags. Clustering was performed using the clean tags,

Table 3Effects of a herbal supplement (HS; mixture of ginsenosides and ginseng polysaccharides) on the growth performance of *P. nigromaculatus*.

Group	IBW/g	FBW/g	WGR/%	SGR/%/d	SR/%
C	12.30 ± 0.07 ^a	32.49 ± 0.52 ^b	164.18 ± 4.90 ^c	1.62 ± 0.03 ^b	92.67 ± 3.33 ^b
T1	12.20 ± 0.04 ^b	33.75 ± 1.32 ^{ab}	176.64 ± 11.83 ^{ab}	1.69 ± 0.07 ^{ab}	97.11 ± 0.39 ^a
T2	12.40 ± 0.05 ^a	35.04 ± 1.75 ^a	182.53 ± 13.18 ^{ab}	1.73 ± 0.08 ^{ab}	98.22 ± 1.02 ^a
T3	12.30 ± 0.04 ^a	35.30 ± 1.30 ^a	187.015 ± 10.31 ^a	1.76 ± 0.06 ^a	97.33 ± 1.33 ^a

Note: IBW, initial body weight; FBW, final body weight; WGR, weight gain rate; SGR, specific growth rate; SR, survival rate. Values are expressed as the mean of six replicates. Values in the same column with different superscript letters differ significantly ($P < 0.05$), the same as the following table.

and chimeric tags were removed. The resulting data were defined as effective tags. Operational taxonomic units were obtained, and abundance of each operational taxonomic unit was calculated using the effective tags [34].

2.7. Bacterial challenge

At the end of the rearing experiment, 30 frogs were randomly selected from each group for bacterial challenge. Each group was divided into three replicates with 10 frogs per replicate. A preliminary experiment was conducted to determine the median lethal dose of *A. hydrophila* against *P. nigromaculatus*, which was found to be 1.5×10^7 CFU/mL. Frogs were intraperitoneally injected with 0.1 mL of a bacterial suspension at the median lethal dose. Mortality in each group was recorded over seven days following the challenge.

2.8. Functional validation of probiotics

In total, 600 *P. nigromaculatus* specimens (initial weight: 12 ± 0.5 g) were selected and divided into control and experimental groups, with three replicates per group and 100 frogs per replicate. Based on the results indicated in section 2.6, probiotic strain (*Lactococcus lactis*) with the most significantly altered gut microbiota was selected for analysis. Cultured bacteria were resuspended in sterilised distilled water to a final concentration of 1×10^7 CFU/mL [35]. The bacterial suspension was evenly sprayed onto the feed daily in the experimental group, whereas the control group was sprayed with sterilised distilled water. The feed ingredients and formulations were the same as those indicated in group C in Table 1. Feeding methods and routine management were performed as described in section 2.2. Body weight and mortality data were recorded for each group at the beginning and end of the experiment. Intestinal samples from the hindgut of three frogs per group were collected and sent to Guangzhou Gene Denovo Biotechnology Co., Ltd. for 16S rRNA sequencing, as described in section 2.6.

2.9. Statistical analyses

Data were analysed using the SPSS software (version 21.0; IBM, USA). Normality was assessed using the Shapiro–Wilk test. Data conforming to a normal distribution and homogeneity of variance were analysed via one-way analysis of variance, followed by Duncan's multiple-range test for post-hoc comparisons. Independent sample *t*-tests were used for pairwise comparisons. Data not meeting the assumptions of normality and homogeneity of variance were analysed using the Kruskal–Wallis non-parametric test. Results are expressed as the mean ± standard deviation, with statistical significance set at $P < 0.05$.

3. Results

3.1. Growth performance

HS significantly enhanced the growth performance of *P. nigromaculatus* (Table 3). WGR and SGR were significantly higher in the T3 group than in the other groups ($P < 0.05$). No significant

Table 4Effects of HS (mixture of ginsenosides and ginseng polysaccharides) on the innate immunity of *P. nigromaculatus*.

Group	C	T1	T2	T3
AST (U/L)	59.10 ± 5.74	54.13 ± 2.48	55.96 ± 11.20	56.39 ± 5.88
ALT (U/L)	55.14 ± 2.29 ^a	51.37 ± 0.97 ^b	56.98 ± 2.85 ^a	52.55 ± 2.39 ^b
AST/ALT	1.07 ± 0.06	1.05 ± 0.03	0.99 ± 0.25	1.08 ± 0.10
SOD (U/mg protein)	56.54 ± 3.34 ^c	66.00 ± 5.17 ^b	81.12 ± 4.49 ^a	49.20 ± 3.74 ^c
CAT (U/mg protein)	10.85 ± 0.30 ^d	21.78 ± 0.12 ^a	16.08 ± 0.08 ^b	12.44 ± 0.24 ^c

Note: AST, alanine aminotransferase; ALT, aspartate transaminase; SOD, superoxide dismutase; CAT, catalase.

differences in WGR and SGR were observed between the T1 and T2 groups ($P > 0.05$). Moreover, SR was significantly higher in all experimental groups than in the control group ($P < 0.05$).

3.2. Non-specific immune indices

No significant difference in AST activity was observed among the groups ($P > 0.05$). ALT activity did not significantly differ between the T2 and control groups ($P > 0.05$) but was significantly lower in the T1 and T3 groups than in the C and T2 groups ($P < 0.05$). Moreover, AST/ALT ratio was approximately 1 in all groups, with no significant difference ($P > 0.05$). With increasing HS inclusion level in the diet, SOD activity initially increased and subsequently decreased. SOD activity was significantly higher in the T2 group than in the other groups ($P < 0.05$); however, no significant difference was observed between the T3 and control groups ($P > 0.05$). CAT activity was significantly higher in all experimental groups than in the control group ($P < 0.05$), with T1 group showing the highest CAT activity ($P < 0.05$), followed by the T2 and T3 groups (Table 4).

3.3. Immunity and intestinal functions

Compared to those in the control group, relative expression levels of zona occludens (*ZO*)-1, *ZO*-2, claudin (*CLD*)-5, and *CLD3* were elevated in all experimental groups. Specifically, *ZO*-2 and *CLD5* levels were significantly higher in the T2 group than in the control group ($P < 0.0005$), and *CLD3* levels were significantly higher in the T3 group than in the control group ($P < 0.0005$). In terms of immunity-related genes, relative *OL-CATH1* levels were significantly higher in the T2 group than in the other groups ($P < 0.0005$). Relative IL10 and IL12 levels were lower in all experimental groups than in the control group. Particularly, IL10 levels were significantly lower in the T1 and T2 groups than in the control group ($P < 0.0005$), and IL12 levels were significantly lower in the T1, T2, and T3 groups than in the control group ($P < 0.0005$). However, no significant difference in relative NF-κB levels was observed between the experimental and control groups ($P > 0.05$; Fig. 1).

3.4. Intestinal flora

As shown in Fig. 2, *Bacteroides*, *Lactococcus*, and

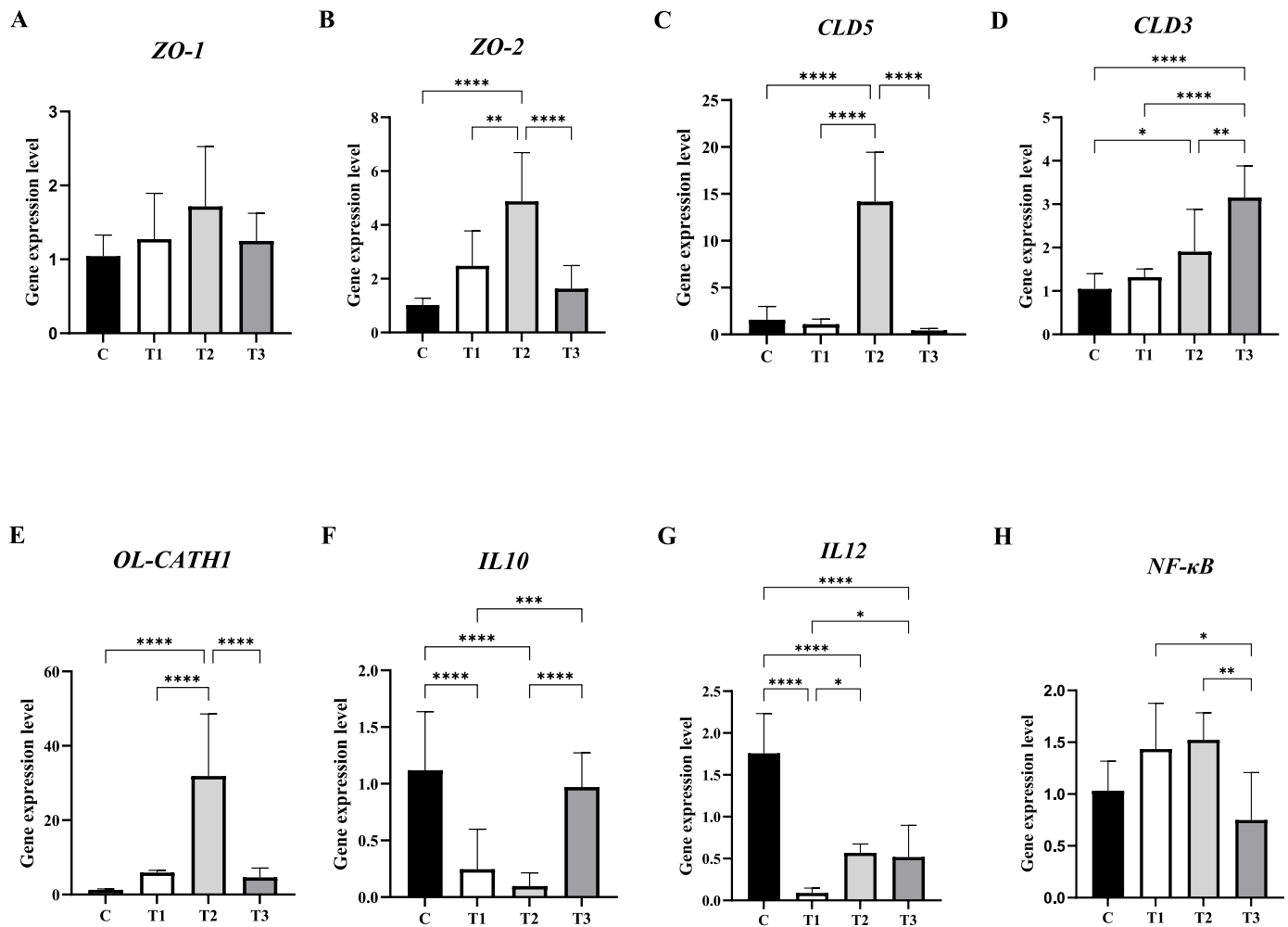


Fig. 1. Changes in the immunity and intestinal functions of *Pelophylax nigromaculatus* after 60-d feeding with herbal supplement (HS; mixture of ginsenosides and ginseng polysaccharides)-supplemented diets. * $P < 0.05$, ** $P < 0.005$, *** $P < 0.0005$, and **** $P < 0.0005$.

Erysipelatoclostridium were the dominant genera in the gut microbiota of *P. nigromaculatus* in all groups. Notable changes were observed in the relative abundances of *Lactococcus* and *Escherichia-Shigella*. Relative abundance of *Lactococcus* was significantly higher in all experimental groups than in the control group, with the T2 group exhibiting the highest abundance ($P < 0.05$), followed by the T1 and T3 groups. In contrast, no significant difference in the relative abundance of *Escherichia-Shigella* was observed between the T2 and control groups ($P > 0.05$). However, relative abundance of *Escherichia-Shigella* was significantly lower in the T1 and T3 groups than in the T2 and control groups ($P < 0.05$).

3.5. *P. nigromaculatus* resistance to *A. hydrophila* infection

SR of *P. nigromaculatus* was assessed seven days after *A. hydrophila* challenge. SR was significantly higher in the experimental groups than in the control group ($P < 0.05$). Among the experimental groups, T2 group exhibited the highest SR, followed by the T1 and T3 groups ($P < 0.05$). Notably, no significant difference in SR was observed between the T1 and T3 groups ($P > 0.05$; Fig. 3).

3.6. Effect of feed sprayed with *L. lactis* on *P. nigromaculatus* growth

To determine whether HS improves the growth performance of *P. nigromaculatus* by increasing the intestinal abundance of *Lactococcus*, *L. lactis* was added to the diet of *P. nigromaculatus* (Table 5). Notably,

SGR, WGR, and SR were higher in the T group than in the C group. Moreover, SR was significantly higher in the T group than in the C group ($P < 0.05$).

3.7. Effects of feed sprayed with *L. lactis* on *P. nigromaculatus* gut microbiota

Fig. 4 shows the top 10 most abundant genera in *P. nigromaculatus* gut microbiota, with others categorised as "Other" or "Unclassified." Beneficial bacteria, including *Bacteroides*, *Cetobacterium*, *Parabacteroides*, and *Romboutsia*, were more abundant, whereas harmful bacteria, such as *Brachyspira* and *Mucispirillum*, were less abundant in the experimental groups. Relative abundance of *Parabacteroides* was significantly higher ($P < 0.05$), whereas that of *Brachyspira* was significantly lower ($P < 0.05$) in the experimental groups than in the control group. Similar to that described in section 3.4, frogs fed HS diets exhibited high abundance, whereas those fed diets sprayed with *L. lactis* exhibited low abundance of *Lactococcus* and were categorised as "Other."

4. Discussion

Ginseng extracts, ginseng polysaccharides, and ginsenosides are commonly used as natural plant-based additives in aquaculture feed due of their ability to enhance growth performance, disease resistance, and metabolism [14]. In this study, growth performance of *P. nigromaculatus*

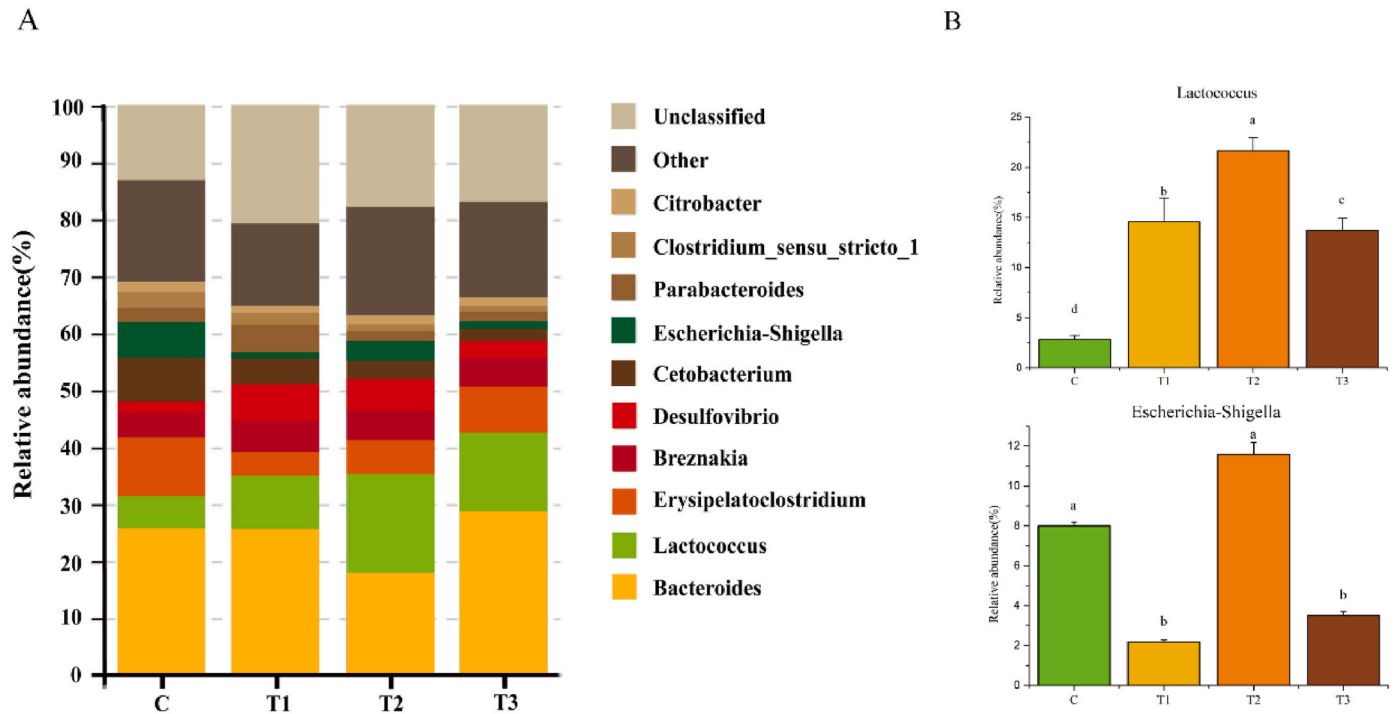


Fig. 2. Effects of HS (mixture of ginsenosides and ginseng polysaccharides) on the gut microbiota of *P. nigromaculatus*. (A) Relative abundance changes in gut microbiota at the genus level. (B) Significantly altered beneficial and harmful genera. Groups with same superscript letters are not significantly different ($P > 0.05$), whereas those with different superscript letters are significantly different ($P < 0.05$).

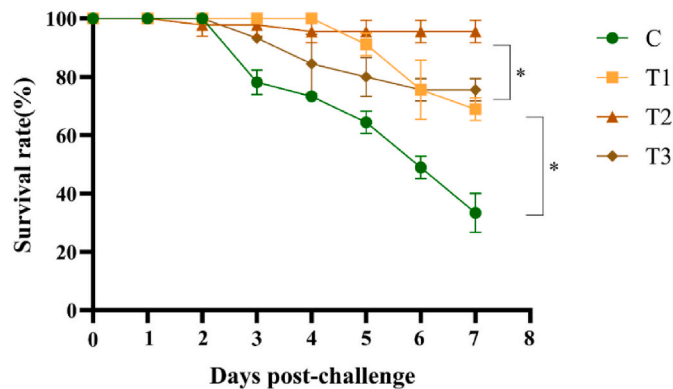


Fig. 3. Survival rate of *Aeromonas hydrophila*-infected *P. nigromaculatus*.

was significantly higher in the experimental groups than in the control group, consistent with a previous report on tilapia (*O. niloticus*) by Li [36]. Furthermore, Mansour [20] revealed that diet supplementation with 0.4 g/kg ginseng extract is more effective in enhancing the plasma testosterone levels and growth performance of Nile tilapia (*O. niloticus*) than that with 0.6 g/kg *Tribulus* extract and 6 g/kg date palm pollen grains. Although 1.2 g/kg *Tribulus* extract exerts superior growth-promoting effects, ginseng extract exhibits good performance at a low supplementation level, showing high cost-effectiveness.

AST and ALT are commonly used indicators of internal organ health, with ALT serving as a specific marker of liver function. AST/ALT ratio

close to 1 is generally indicative of normal liver function [37]. Here, AST activity was not significantly different among the groups ($P > 0.05$), whereas ALT activity was significantly lower in the T1 and T3 groups than in the control group. However, AST/ALT ratio remained close to 1 in all groups, with no significant difference ($P > 0.05$), suggesting that dietary supplementation with an appropriate level of HS does not negatively affect the liver, heart, or muscle tissue health in *P. nigromaculatus*. As the supplementation level of HS increased, serum SOD activity initially increased and then decreased, peaking in the T2 group. CAT activity was significantly higher in all experimental groups than in the control group ($P < 0.05$). Both SOD and CAT are essential antioxidant enzymes; SOD primarily reflects the acute immune responses, whereas CAT represents the overall antioxidant defence capacity of the organism [38]. OL-CATH1, encoding an antimicrobial peptide, is a critical component of the innate immune system, directly targeting and disrupting the cell membranes of bacterial and fungal pathogens [39]. *NF- κ B*, a central regulator of inflammatory responses, showed no significant difference in relative expression between the experimental and control groups ($P > 0.05$), suggesting that the systemic inflammatory responses were not significantly enhanced [40]. However, relative expression of *OL-CATH1* was significantly upregulated, suggesting that the enhancement of immune defence was primarily achieved via non-inflammatory innate immune pathways [41]. Furthermore, significant downregulation of *IL-10* and *IL-12* levels in the HS-treated groups highlighted the potential role of HS in maintaining immune homeostasis. *IL-10* is a key anti-inflammatory cytokine suppressing excessive immune responses, whereas *IL-12* primarily promotes Th1-type adaptive immunity and pro-inflammatory cytokine

Table 5

Effect of feed sprayed with *Lactococcus lactis* on the growth performance of *P. nigromaculatus*.

Group	IBW/g	FBW/g	WGR/%	SGR/%/d	SR/%
C	12.30 $\pm$ 0.05	35.31 $\pm$ 1.03	187.05 $\pm$ 7.21	1.76 $\pm$ 0.04	70.67 $\pm$ 0.06
T	12.30 $\pm$ 0.04	37.58 $\pm$ 1.72	205.50 $\pm$ 13.00	1.86 $\pm$ 0.07	81.33 $\pm$ 0.16 ^a

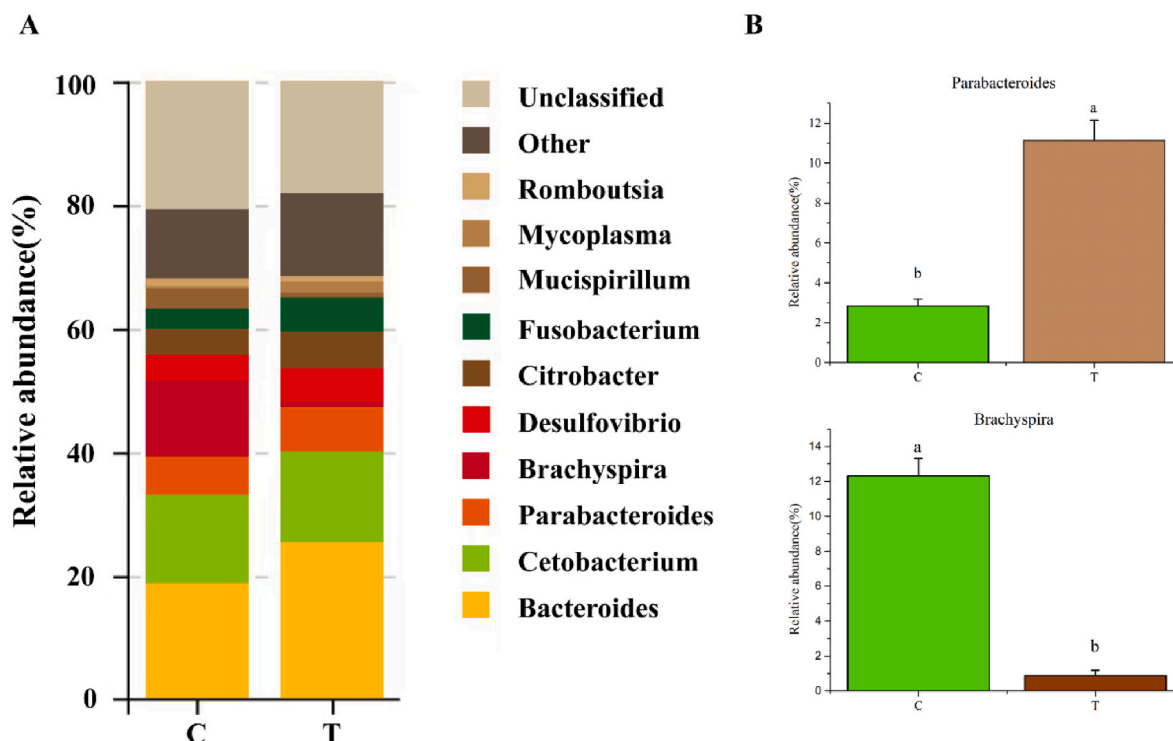


Fig. 4. Effects of feed sprayed with *Lactococcus lactis* on the gut microbiota of *P. nigromaculatus*. (A) *L. lactis*-induced changes in gut microbiota at the genus level. (B) Significantly altered beneficial and harmful genera.

production. Concurrent reduction of both cytokine levels suggests that HS mitigates excess inflammatory responses and prevents the over-activation of the adaptive immune system [42]. Combined with the significant increase in SOD and CAT activities in the experimental groups, these results suggest that HS enhances both the innate immunity and antioxidant defence responses, establishing a protective barrier improving the overall disease resistance of *P. nigromaculatus*. Ginsenosides in HS, as the primary active component, promote the synthesis of antimicrobial peptides by activating the Toll-like receptor signalling pathways [43]. Moreover, ginseng polysaccharides effectively scavenge free radicals, whereas trace elements and vitamins act synergistically to enhance the antioxidant capacity [44,45].

Intestinal genes *ZO-1*, *ZO-2*, *CLD3*, and *CLD5* are tight junction-related genes playing critical roles in maintaining the intestinal barrier function and tissue integrity [46]. This study revealed elevated *ZO-1*, *ZO-2*, *CLD3*, and *CLD5* levels in the experimental groups. Specifically, *ZO-2* and *CLD5* levels were significantly higher in the T2 group than in the control group. Our findings suggest that HS effectively regulates the intestinal permeability and enhances the intestinal barrier function in *P. nigromaculatus*. Gut microbiota analysis revealed that the relative abundance of *Lactococcus* was significantly higher in the experimental groups than in the control group, with T2 showing the highest abundance. In contrast, relative abundance of *Escherichia-shigella* was significantly low in the T1 and T3 groups. These results suggest that HS promotes tight junction protein expression by improving the intestinal microbial environment. HS contains ginseng polysaccharides that act as prebiotics and serve as a nutrient source for beneficial gut microbes, promoting their proliferation. Additionally, secondary metabolites derived from the microbial transformation of ginsenosides in the gut further enhance the growth of beneficial bacteria [47].

To verify whether the most abundant *Lactobacillus* species plays a key role in regulating the intestinal health of *P. nigromaculatus*, a validation experiment was performed, in which *L. lactis* was sprayed onto the feed and fed to the frogs. The results revealed significantly increased relative abundance of the beneficial bacterium, *Parabacteroides*, and

significantly decreased relative abundance of the harmful bacterium, *Brachyspira*, along with notable improvement in growth performance in the experimental groups. However, relative abundance of *Lactobacillus* in the frog intestines remained low across all groups. *Lactobacillus* is not a dominant member of the gut microbiota in fish, and its abundance is influenced by environmental and nutritional factors [48]. Xu [49] reported that feeding grass carp (*Ctenopharyngodon idella*) juveniles with feed supplemented with *Lactobacillus pentosus* R1 increases *Lactobacillus* abundance in the gastrointestinal tract. However, R1 was not a dominant gut microbe, as its composition in the gut remained low even after significant consumption. In aquaculture, probiotics are typically administered via feed sprays or water supplementation. As probiotics cannot permanently colonise the gastrointestinal tract, continuous supplementation is essential for their effective function [50]. Different administration methods significantly affect the abundance and colonisation efficiency of probiotics in the gastrointestinal tract of aquatic animals. According to Ventura [51], *Bifidobacterium* exhibits host-specific colonisation mechanisms across various hosts while maintaining a relatively conserved core colonisation strategy in mammals. In this study, lactic acid bacterial strain was originally derived from humans, which possibly affected its long-term colonisation in the gastrointestinal tract of *P. nigromaculatus*. However, short-term stimulation with *L. lactis* significantly improved the gut microbial environment of *P. nigromaculatus*. All experimental groups exhibited a relatively high abundance of *Lactobacillus*, possibly due to the regulatory effects of HS on the gut microbiota. Ginseng polysaccharides in HS significantly promote the proliferation of beneficial bacteria, such as *Lactobacillus* and *Bifidobacterium* [52]; this was further confirmed by our validation experiment.

Healthy gut microbiota environment facilitates nutrient absorption, and gut microbiota interacts with the intestinal immune system to enhance the local and systemic immunity [53]. In this study, relative abundance of beneficial gut bacteria was significantly higher in the experimental groups than in the control group, with the T2 group showing the highest abundance. Additionally, high antimicrobial

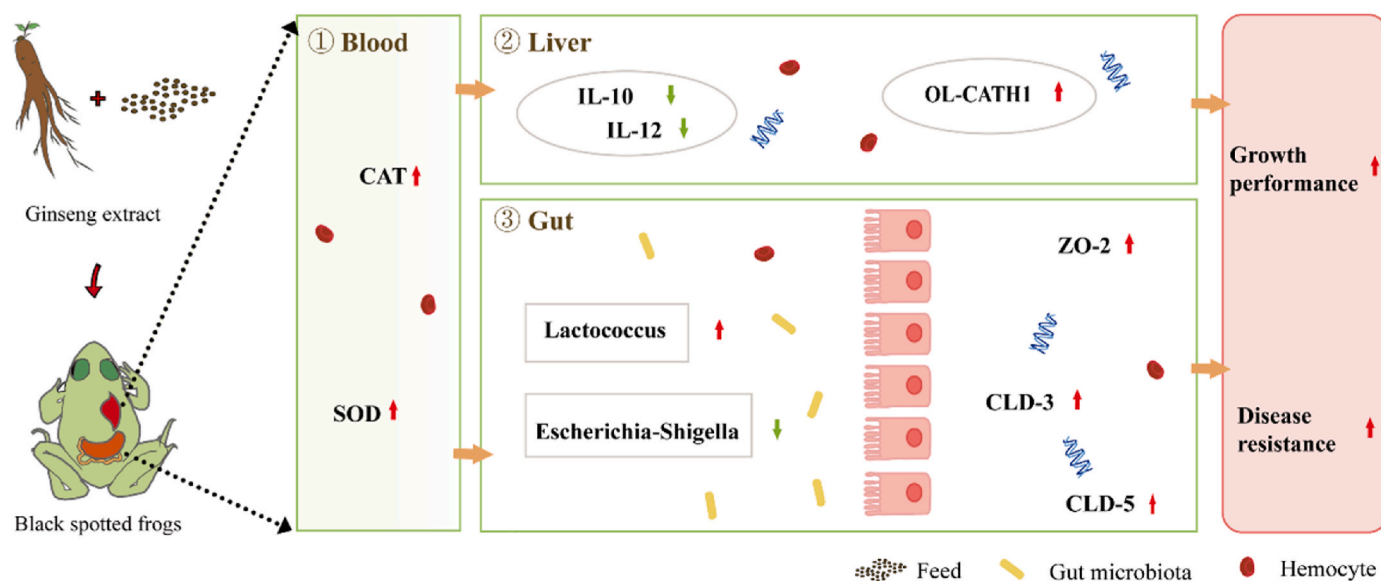


Fig. 5. Summary diagram of the main study findings.

peptide levels and antioxidant capacity further improved the disease resistance of *P. nigromaculatus*. When *P. nigromaculatus* was challenged with *Aeromonas hydrophila*, SR was significantly higher in all experimental groups than in the control group, with the T2 group showing the highest SR. Combined with the growth experiment results, these findings confirmed that HS significantly improved both the growth performance and disease resistance of *P. nigromaculatus*. Crude extract of *Phyllanthus urinaria* (10,000 ppm) significantly improves the growth and feed efficiency, but not SR, of *Litopenaeus vannamei*. Moreover, 2 % *Aloysia citrodora* enhances the immunity of *Oncorhynchus mykiss*, without affecting its growth [54]. Compared to other plant-based feed additives, HS exhibited superior overall utility. As a natural immunomodulator, it poses no risk of antibiotic residues or development of antimicrobial resistance, serving as an effective alternative antibiotic for *P. nigromaculatus* farming. Our experimental results suggest that continuous HS supplementation does not damage the internal organs, including the liver. However, its potential long-term risks, such as bioaccumulation, gut microbiota imbalance, and unintended resistance effects, warrant further investigation. Future studies should assess the chronic toxicity, safe dosage range, and optimal application cycles of HS to facilitate its widespread use in large-scale aquaculture.

5. Conclusions

1. HS enhanced the growth performance and immunity of *P. nigromaculatus* by improving its gut microbiota environment.
2. HS enhanced the disease resistance of *P. nigromaculatus* by increasing its antimicrobial peptide activity and antioxidant capacity.
3. Our findings suggest diet supplementation with 0.04 % HS for efficient *P. nigromaculatus* aquaculture (Fig. 5).

Author contributions

Yazhou Hu: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft. Yixin Liang: Formal analysis, Visualization, Data curation, Writing – original draft. Le Han: Investigation, Resources. Denghui Hu: Software, Validation. Ming Li: Visualization, Data curation. Zhaobin Zhou: Resources, Project administration. Yi Hu: Methodology, Formal analysis. Xiaoqing Wang: Investigation, Data curation. Yulong Yin: Conceptualization, Supervision, Writing – review & editing, Funding acquisition.

Ethical standards

Animal experiments complied with the regulations of the Animal Care and Use Committee of the College of Fisheries, Hunan Agricultural University (Changsha, China; Approval Number: 20220316; Approval Date: March 16th, 2022).

Declaration of generative AI and AI-assisted technologies in the writing process

This article utilized AI-assisted technologies, including ChatGPT, to enhance the language quality and readability of the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fsi.2025.110553>.

Data availability

Data will be made available on request.

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